10.1 $U=\frac{k(Q+q)}{b}-\frac{kq}{a}+\frac{kq}{r}$

$=\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{b}-\frac{1}{a}+\frac{1}{r}\right)+\frac{Q}{4\pi\varepsilon_0 b}$

10.4. (1) $U=\frac{kq}{l}$

(2). $\frac{kq'}{R}+\frac{kq}{l}=0.$ $q'=\frac{R}{-l}q$

10.6. (1) $\begin{cases}\frac{kq_1}{R_2}+\frac{kq_2}{R_3}+\frac{kq}{R_2}=\frac{kq_1}{R_3}+\frac{kq_2}{R_3}+\frac{kq}{R_3}\\ Q=q_1+q_2\end{cases}$

可得 $\begin{cases}q_1=-q\\ q_2=Q+q.\end{cases}$

$U_1=\frac{kq}{R_1}+\frac{k(-q)}{R_2}+\frac{k(Q+q)}{R_3}$

$=\frac{(R_2R_3-R_1R_3+R_1R_2)q+R_1R_2Q}{4\pi\varepsilon_0R_1R_2R_3}$

$U_2=\frac{k(-q)}{R_2}+\frac{k(Q+q)}{R_3}+\frac{kq}{R_2}=\frac{Q+q}{4\pi\varepsilon_0R_3}$

(2). $U_1=U_2=\frac{k(Q+q)}{R_3}=\frac{Q+q}{4\pi\varepsilon_0R_3}$

(3). $U_1=0.$ $\frac{k(Q+q')}{R^3}-\frac{kq'}{R_2}+\frac{kq'}{R_1}=0.$ 得 $q'=\frac{R_1R_2Q}{-R_1R_2-R_2R_3+R_1R_3}$

$U_2=\frac{kq'}{R_2}-\frac{kq'}{R_2}+\frac{k(Q+q')}{R_3}=\frac{k(-R_2R_3Q+R_1R_3Q)}{(-R_1R_2-R_2R_3+R_1R_3)R_3}$

$=\frac{(R_2-R_1)Q}{4\pi\varepsilon_0(R_1R_2+R_2R_3-R_1R_3)}$

10.11 (1). $\begin{cases} \dfrac{kq_A}{a} = \dfrac{kq_B}{b} \\ q_A + q_B = Q \end{cases}$ 得 $\begin{cases} q_A = \dfrac{a}{a+b}Q \\ q_B = \dfrac{b}{a+b}Q \end{cases}$

(2). $Q = CU$, $U = \dfrac{kq_A}{a} = \dfrac{kQ}{a+b}$ 相当于电容并联.

$$C = \frac{Q}{U} = \frac{Q}{\frac{kQ}{a+b}} = \frac{a+b}{4\pi\varepsilon_0}$$

10.13

$d \gg r$.

$$E \cdot 2\pi x = \frac{\lambda}{\varepsilon_0}$$

$$E_{总} = 2 \cdot \frac{\lambda}{2\pi x \varepsilon_0} = \frac{\lambda}{\pi x \varepsilon_0}$$

$$U = \int_r^d \frac{\lambda}{\pi x \varepsilon_0} dx = \frac{\lambda}{\pi\varepsilon_0} \ln x \Big|_r^d = \frac{\lambda}{\pi\varepsilon_0} \ln \frac{d}{r}$$

$$C = \frac{Q}{U} = \frac{\lambda}{\frac{\lambda}{\pi\varepsilon_0} \ln \frac{d}{r}} = \frac{\pi\varepsilon_0}{\ln \frac{d}{r}}$$

10.14. $E \cdot 4\pi x^2 = \dfrac{Q}{\varepsilon_0}$, $U = \displaystyle\int_a^b \frac{Q}{\varepsilon_0} \frac{}{4\pi x^2} dx = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$.

则 $Q = \dfrac{4\pi\varepsilon_0 U}{\frac{1}{a} - \frac{1}{b}}$

$$E = \frac{\sigma}{\varepsilon_0} = \frac{\frac{Q}{4\pi a^2}}{\varepsilon_0} = \frac{U}{\left(\frac{1}{a} - \frac{1}{b}\right)a^2} = \frac{U}{a - \frac{a^2}{b}}$$

当 $a = \frac{b}{2}$ 时, $E_{min} = \dfrac{4U}{b}$